

Avi Vyas

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PROFESSIONAL EXPERIENCE

Software Engineer (Backend Engineer)

07/2025 – Present | Noida, India

To The New (TTN) 🔗

- Developed and maintained scalable backend microservices using **Java** and **Spring Boot** for a platform serving **18M+ users**.
- Optimized **Elasticsearch-based search and autocomplete APIs**, reducing response time from **1s+ to ~80ms** while achieving **1,500+ TPS** with **~50% of production infrastructure**, significantly improving search performance, throughput, and infrastructure efficiency.
- **Developing an AI-powered Root Cause Analysis (RCA) microservice** that detects **KPI threshold breaches** and generates **evidence-based RCA** using application code, logs, and runtime data, with follow-up analysis capabilities, designed for adoption across **multiple teams and projects**.
- Regularly contribute to **production performance optimization and incident RCA**, investigating alerts, bottlenecks, and service-level issues across microservices and implementing targeted improvements to **reliability, latency, and system efficiency**.
- Developed an automated **free channel upgrade** solution enabling eligible subscribers to upgrade to **HD channels** at the same subscription price across multiple touchpoints, later generalized it into a **reusable microservice** adopted by multiple teams, eliminating duplicate implementations.
- Architected and developed **configurable promotions** across multiple implementation phases, supporting recurring and multiple promotion types through **business-rule-driven configuration**, reducing dependency on code changes for new promotional campaigns.

SKILLS

Languages — Java (Primary), Python

Frameworks & AI — Spring Boot 4, Spring AI, Spring Security, Spring Data JPA, Hibernate, MCP, GitHub Spec Kit, Cursor, FastAPI

Databases & Messaging — PostgreSQL, MySQL, Redis, Elasticsearch, RabbitMQ

Cloud & DevOps — AWS (EC2, S3), Docker, Git, GitLab, Maven, Gradle

Testing & Core — JUnit, Mockito, SLF4J, Microservices, REST APIs, Multithreading, OOP, SOLID, Data Structures

PROJECTS

E-Commerce Backend System 🔗

Java, Spring Boot, MySQL, Spring Security, JWT

- Developed a **production-style e-commerce backend with 50+ REST APIs** supporting Admin, Seller, and Customer workflows with **JWT-based authentication and role-based authorization**.
- Designed REST APIs for **product, catalog, and user management**, including flexible metadata-driven product structures.
- Implemented **pagination, validation, exception handling, and layered architecture**, with a focus on maintainability and scalable API design

Doctor Appointment System 🔗

Python, FastAPI, MySQL, SQLAlchemy, JWT

- Built a **FastAPI-based appointment management system** with JWT authentication and role-based access control for doctors and patients.
- Designed **time-slot scheduling with conflict detection** and appointment lifecycle management.
- Implemented database persistence using **SQLAlchemy and MySQL** with modular API/service architecture.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

2021 – 2025

BML Munjal University

CGPA: 8.14 / 10